

Alvaro Perdomo Gutierrez

Senior Full Stack Engineer

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with **12+** years across fintech, healthcare, and e-commerce, building production-grade systems in Python and React with cloud-native architecture. Expert in FastAPI, PostgreSQL, and AWS infrastructure including ECS Fargate and Aurora. Experienced designing end-to-end features, optimizing data pipelines, and leveraging AI-driven development practices to accelerate delivery and improve system reliability.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Backend: FastAPI, Django, Node.js, Express

Frontend: React, Next.js, Redux, Tailwind CSS

Database: PostgreSQL, Aurora, Redis, DynamoDB, MongoDB

Testing: Jest, Pytest, Cypress, Playwright

Cloud & DevOps: AWS, ECS Fargate, AWS CDK, Terraform, Docker, Kubernetes, GitHub Actions, CloudWatch, S3, Lambda

Others: RESTful API, GraphQL, Cursor, GitHub Copilot, LLM Integration, Git, Jira, Confluence

EXPERIENCE

Leanware

May 2024–Present

Senior Software Engineer

- Led the end-to-end development of an AI-powered SaaS platform for algorithmic trading, designing **Python** and **FastAPI** backend services that handled real-time market data ingestion and portfolio optimization, integrating with external trading APIs for seamless order execution.
- Architected and deployed cloud-native infrastructure on **AWS** using **ECS Fargate** and **Aurora PostgreSQL**, establishing automated CI/CD pipelines with **GitHub Actions** that reduced deployment time from 35 minutes to under 10 minutes and improved system reliability across staging and production environments.
- Collaborated with product and data teams to deliver AI-driven development features using **GitHub Copilot** and **LLM** integration for code generation and debugging, accelerating feature velocity by approximately 25% while maintaining code quality standards through rigorous peer reviews.
- Optimized **PostgreSQL** schema design and query performance for high-throughput financial transactions, introducing indexing strategies and query rewrites that reduced average query latency from 800ms to 310ms, enabling the platform to handle 40k+ daily trading operations without degradation.

- Implemented **Redis**-backed session caching and distributed rate limiting to protect trading endpoints from abuse, eliminating unauthorized access attempts and reducing database read load by 45% during peak trading hours.
- Contributed to frontend feature development using **TypeScript** and **React**, building responsive dashboards for real-time portfolio visualization and trade execution, ensuring cross-browser compatibility and sub-500ms page load times.
- Drove continuous improvements in developer experience by establishing Infrastructure as Code practices with **AWS CDK**, creating reusable constructs that enabled team members to provision new services without manual configuration, reducing environment setup time from 2 days to 30 minutes.
- Participated in on-call rotation and incident response, resolving production issues within service-level agreements and implementing monitoring dashboards with **CloudWatch** metrics to provide visibility into application health and performance bottlenecks.
- Mentored two junior engineers on full-stack development practices, conducting code reviews focused on error handling, edge case coverage, and AWS best practices, helping them successfully ship three independent features to production.
- Established pragmatic testing strategies by introducing **Pytest** for backend unit tests and **Cypress** for critical user workflows, achieving 72% code coverage and reducing regression bugs in production by 30%.
- Evaluated and adopted AI-assisted development tools including **Cursor** for rapid prototyping, applying LLM-based suggestions for routine tasks like boilerplate generation and schema migrations while maintaining careful review of generated code for security and correctness.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Spearheaded the integration of Lily AI computer vision system into the resale platform using **Python** and **FastAPI**, processing 275k+ product images weekly and delivering 15% increase in sell-through and 2.8% iOS conversion lift, directly contributing to \$18M revenue impact.
- Drove migration of legacy ML model serving infrastructure to **Databricks** platform, reducing model deployment time from 6 months to 1 month by leveraging **LLM**-enabled notebooks and standardized feature engineering pipelines, accelerating MVP iterations from 8 weeks to 2 weeks.
- Optimized **PostgreSQL** queries across the product catalog system serving 172M+ items, introducing materialized views and composite indexing that cut catalog query latency from 1.2s to 280ms, enabling real-time personalization features for 2.5M active members.
- Architected **AWS** infrastructure for image processing pipelines using **ECS Fargate** and **Lambda** functions, processing 100k+ auto-generated product photos daily with automatic quality validation, reducing manual photography work by 60% and eliminating processing bottlenecks.
- Built Image Search and Style Chat features using **React** and **TypeScript** frontend components, integrating embeddings-based vector search with **Redis** caching to deliver sub-500ms search response times for millions of product queries.

- Implemented **Aurora PostgreSQL** schema redesign to support single-SKU personalization, normalizing previously denormalized data structures that had caused query performance degradation, reducing database CPU usage by 35% during peak traffic windows.
- Collaborated with data science team on ML pipeline orchestration, establishing **GraphQL** API layer between backend services and model serving infrastructure, reducing inter-service communication latency and enabling real-time model serving for dynamic pricing algorithms.
- Led code review and quality initiatives using **GitHub Actions** CI/CD, establishing automated testing gates that caught 92% of regressions before production deployment and reduced post-release incident rate by 40%.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Played a key role in scaling the precision-matching ML platform from supporting 300k to 2M+ members, architecting **Python**-based service enhancements that preserved clinical matching validity while accommodating 6x user growth without additional infrastructure scaling.
- Designed and launched the Care Navigation platform globally across 12 countries, implementing localization features using **React** and **TypeScript** and jurisdiction-aware matching logic in **FastAPI** backends, enabling dependent eligibility determination and country-specific compliance workflows.
- Engineered HIPAA-compliant **AWS** infrastructure for Fortune 500 enterprise onboarding (PepsiCo, Instacart, Bain), configuring **Aurora PostgreSQL** databases with encryption at rest, implementing fine-grained **Redis**-backed session isolation, and achieving SOC 2 Type II certification.
- Optimized member matching performance by refactoring database queries against 5M+ clinical records, replacing N+1 queries with intelligent **PostgreSQL** window functions that reduced matching latency from 2.3s to 380ms, enabling real-time therapist assignment.
- Built operational ML features for dropout-risk prediction and demand forecasting, creating data pipelines in **Python** that ingested behavioral signals, trained gradient boosting models, and surfaced predictions via **GraphQL** API to care coordination teams.
- Contributed to infrastructure improvements using **Terraform** for reproducible environment provisioning, establishing staging parity with production and enabling parallel testing of schema migrations, reducing deployment risk for 50+ database changes annually.
- Participated in code reviews and technical discussions focused on system reliability, advocating for defensive coding practices and comprehensive error handling that reduced production incidents from 8 per week to 2 per week during my tenure.

Tencent Games

Nov 2014–Nov 2020

Game Server Engineer

- Built backend services for Honor of Kings, a top mobile MOBA that grew to 100M+ DAU by 2020, developing server logic in **Python** and **C++** for match orchestration, rank calculation, and real-time player messaging, handling peak concurrent load of 2M+ simultaneous players.

- Architected distributed game state persistence layer using **TcaplusDB** (Tencent's proprietary NoSQL database), supporting 40M+ daily active operations with sub-100ms latency guarantees, ensuring consistent player progression across Asia-Pacific regions.
- Implemented anti-cheat infrastructure powered by the ACE kernel-level detection system, reducing account compromise incidents by 78% through behavioral anomaly detection and real-time ban enforcement across millions of concurrent players.
- Contributed to PUBG Mobile backend systems on **Unreal Engine 4**, designing server-side validation and anti-cheat integration for 100-player battle royale matches, supporting launch across 20+ countries and achieving 50M+ registered players within 12 months.
- Optimized network communication protocols for mobile gameplay, reducing average packet size by 35% through message compression and delta encoding, improving gameplay fluidity for players on 3G networks and reducing server bandwidth costs by \$200k annually.
- Collaborated with GCloud platform team on infrastructure scaling, establishing real-time monitoring dashboards using custom telemetry and supporting rapid expansion from 20M to 100M+ DAU with zero gameplay disruptions during scaling transitions.
- Mentored junior backend engineers on distributed systems design, focusing on fault tolerance, eventual consistency patterns, and debugging production issues in high-concurrency environments, with three mentees promoted to senior roles during my tenure.

EDUCATION

The Hong Kong Polytechnic University
Bachelor of Science in Computer Science

Sep 2010–Nov 2014